University San Francisco de Quito

Artificial Intelligence

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Class Activity - Writing a Genetic Algorithm

Provide a short explanation of the algorithm and the choices it made through its execution.

This genetic algorithm searches for the minimum of $f(x, y) = (x + 2)^2 + (y - 2)^2 + 10$ by evolving a population of bit-string solutions. The true minimum is at $(-2, 0)$ with $f = 10$.

For the representation, each individual is a 16-bit chromosome: 8 bits x for and 8 for y. The first bit of each half is the sign. The remaining 7 bits are a magnitude scaled by 10, so each variable lives in about $[-12.7, 12.7]$. Fitness is just f of the decoded (x, y), where lower is better.

Each epoch starts by ranking the population of 100 by fitness, then elitism copies the top 10 unchanged into the next generation so the current best can't be lost. The other 90 individuals are made by picking two elites at random, mixing them with single-point crossover, then applying uniform bit-flip mutation, where each bit flips independently with probability 1/16. That combination is intentional: crossover recombines good fragments, mutation adds small random exploration, and elitism keeps the search from sliding backward.

The random start was already near the bowl, around $(-1.5, 2.0)$. with $f = 10.25$. Over the next few epochs, crossover and mutation nudged the elite toward the bottom: (−2.3, 2.1), then (−1.9, 2.0), then (−2.0, 2.0). By epoch 7 the algorithm had the exact minimum $f = 10$. After that the plot is flat: elitism keeps that solution, so later mutations only perturb copies while the best fitness stays at 10 for the remaining epochs.